

Power Query Exercise 1

Getting Started with Power BI & Power Query

Data Analytics | Power BI Desktop | Beginner Level

Duration	Level	Tool	Dataset
30-45 min	Beginner	Power BI Desktop	Retail Sales Q1

1. Overview

In this exercise you will take your first steps with Microsoft Power BI Desktop and its built-in data transformation tool, Power Query. You will load a real dataset, explore the Power Query Editor interface, and carry out five essential transformations that form the foundation of every data analytics workflow.

What You Will Learn

By the end of this exercise you will be able to:

- Launch Power BI Desktop and connect to a CSV data file
- Navigate the Power Query Editor interface
- Rename a query/table to a meaningful name
- Remove columns that are not needed for analysis
- Rename columns to clearer, more descriptive names
- Change a column's data type to the correct type
- Add a calculated column using a simple formula
- Load the transformed data into the Power BI data model

The Dataset — Retail Sales Q1 2024

You will work with `Retail_Sales_Q1_2024.csv` — a fictional retail sales dataset covering January through March 2024. It contains 50 transactions across three product categories sold by five sales representatives.

Column	Data Type	Description
TransactionID	Text	Unique identifier for each sale (e.g. TXN-1001)
Date	Text*	Date the sale was made (MM/DD/YYYY)

CustomerName	Text	Name of the purchasing company
City	Text	Customer city
State	Text	Customer state abbreviation
Category	Text	Product category (Electronics / Office Supplies / Furniture)
Product	Text	Product name
Quantity	Text*	Number of units sold
UnitPrice	Text*	Price per unit in USD
Discount	Text*	Discount rate (0.00 – 0.25)
SalesRep	Text	Name of the sales representative
Region	Text	Sales region
Notes	Text	Optional order notes — not needed for analysis

** These columns will need their data types corrected — one of your tasks in this exercise.*

2. Launching Power BI Desktop

Step 1 — Open Power BI Desktop

1. Click the Windows Start menu and type Power BI Desktop.
2. Click the app to launch it.
3. On the splash screen click Get data. If you see the Home screen instead, click Get data on the Home ribbon.

Step 2 — Connect to the CSV File

Power BI can connect to many data sources. In this exercise you will load a CSV file.

1. In the Get data window, select Text/CSV from the list (or type csv in the search box). Click Connect.
2. In the file browser, navigate to the folder where you saved Retail_Sales_Q1_2024.csv and select it.
3. Click Open. A preview window appears showing the first few rows of data.
4. Do NOT click Load. Instead, click Transform Data. This opens the Power Query Editor where you will clean and shape the data.

Load vs. Transform Data

Load — brings the data straight into Power BI with no changes. Use this only when data is already clean.

Transform Data — opens Power Query Editor so you can clean, reshape, and enrich data before loading.

For this exercise always choose Transform Data.

3. The Power Query Editor Interface

Before making any changes, take a moment to get familiar with the Power Query Editor layout.

Area	What It Does
Queries Pane (left)	Lists all loaded queries/tables. Click a query name to select it.
Data Preview (centre)	Shows a sample of your data rows and columns.
Applied Steps (right)	Records every transformation you apply — like a history log. You can click any step to review or delete it.
Ribbons (top)	Home, Transform, Add Column, View — organised by task type.
Formula Bar	Shows the M formula for the selected Applied Step.

Key Principle: Non-Destructive Editing

Power Query never changes your original CSV file.

Every transformation is recorded as a step in the Applied Steps pane.

You can delete or edit any step at any time without losing your source data.

4. Task 1 — Rename the Query

When Power BI loads a CSV file it uses the file name as the query name. A good query name is short, descriptive, and has no spaces or special characters. This makes it easier to reference in formulas and reports later.

Instructions

1. In the Queries pane on the left, find the query named Retail_Sales_Q1_2024 (the file name without the .csv extension).
2. Double-click the query name. The name becomes editable.
3. Delete the current name and type: SalesData

4. Press Enter to confirm.

Naming Conventions

No spaces — use CamelCase (SalesData) or underscores (Sales_Data).

Be descriptive but brief — SalesData is better than Data or Retail_Sales_Q1_2024_Final.

Avoid starting names with a number — Power BI will flag this as an error.

Expected Result

The Queries pane now shows SalesData instead of Retail_Sales_Q1_2024.

The Applied Steps pane shows: Source, Promoted Headers, Changed Type.

5. Task 2 — Remove Columns

Not every column in a dataset is useful for analysis. Removing unnecessary columns keeps your data model lean and your reports uncluttered. In this task you will remove two columns that add no analytical value.

Columns to Remove

Column to Remove	Reason
Notes	Free-text order notes — inconsistent and not used in any calculation or filter.
State	Customer state is already implied by City; we will use Region for geographic analysis.

Instructions — Method 1: Right-click

1. In the data preview, click the Notes column header to select it.
2. Right-click the column header and choose Remove.
3. The column disappears and a Removed Columns step appears in Applied Steps.

Instructions — Method 2: Select Multiple + Remove

1. Click the State column header.
2. The column is highlighted. Right-click and choose Remove.
3. Verify that both Notes and State are gone from the data preview.

Tip — Remove Other Columns

If you want to KEEP only a specific set of columns, right-click any selected column and choose Remove Other Columns instead.

This is useful when a table has many columns and you only need a few.

Expected Result

The data preview now has 11 columns: TransactionID, Date, CustomerName, City, Category, Product, Quantity, UnitPrice, Discount, SalesRep, Region.

Applied Steps shows a new Removed Columns step.

6. Task 3 — Rename Columns

Clear, human-readable column names make your reports easier to understand. Rename the columns in the table below to match the Target Name.

Current Name	Target Name	Why
TransactionID	Transaction ID	Adds a space for readability
CustomerName	Customer Name	Adds a space for readability
UnitPrice	Unit Price	Adds a space for readability
SalesRep	Sales Rep	Shorter, friendlier label

Instructions

1. In the data preview, double-click the TransactionID column header. The name becomes editable.
2. Type Transaction ID and press Enter.
3. Repeat for CustomerName → Customer Name, UnitPrice → Unit Price, SalesRep → Sales Rep.
4. Verify all four columns are renamed in the data preview.

Alternative Method

You can also rename a column by: selecting it → going to the Transform ribbon → clicking Rename.

Both methods produce the same Applied Step.

Expected Result

Four columns are renamed. Applied Steps shows Renamed Columns (one step per rename, or one combined step).

The header row now reads: Transaction ID | Date | Customer Name | City | Category | Product | Quantity | Unit Price | Discount | Sales Rep | Region

7. Task 4 — Change Column Data Types

A data type tells Power BI how to treat a column's values — as text, a number, a date, and so on. When you load a CSV file, Power BI guesses types automatically but often gets them wrong. Correct data types are essential for accurate calculations and sorting.

Data Types You Will Use

Data Type	Use When...
Text	Values are labels, codes, or names — not for arithmetic.
Whole Number	Values are integers (counts, IDs, quantities).
Decimal Number	Values have decimal places (prices, rates, percentages).
Date	Values represent calendar dates.
True/False	Values represent a yes/no or on/off condition.

Columns to Fix

Column	Change To	Why
Date	Date	Enables date filters and time-intelligence in reports.
Quantity	Whole Number	Quantities are integers; arithmetic must work on them.
Unit Price	Decimal Number	Prices have cents; decimal precision is required.
Discount	Decimal Number	Discount rate is a decimal (e.g. 0.10 = 10%).

Instructions

1. Click the Date column header to select it.
2. Look at the icon to the LEFT of the column name in the header. It currently shows ABC (Text). Click it — a type menu appears.

3. Select Date. A Changed Type step is added to Applied Steps.
4. Click the Quantity column header. Click the type icon (ABC) and select Whole Number.
5. Click the Unit Price column header. Click the type icon and select Decimal Number.
6. Click the Discount column header. Click the type icon and select Decimal Number.

What the Type Icons Mean

ABC → Text

123 → Whole Number

1.2 → Decimal Number

 → Date

If you accidentally set the wrong type, click the step in Applied Steps and delete it, then try again.

Expected Result

Date column icon changes to a calendar symbol.

Quantity icon changes to 123.

Unit Price and Discount icons change to 1.2.

Applied Steps shows a Changed Type step (may be combined with the earlier auto-detect step).

No red error flags should appear in any column — if they do, check that the source values match the type you selected.

8. Task 5 — Add a Calculated Column

A calculated column (called a Custom Column in Power Query) lets you derive a new value from existing columns using a formula. You will add two new columns to make your dataset analysis-ready.

Task 5A — Add Sale Amount

Sale Amount is the revenue for each transaction after applying the discount:

Formula

`Sale Amount = Quantity × Unit Price × (1 - Discount)`

1. In the Add Column ribbon at the top, click Custom Column.
2. In the New column name box, delete the default text and type: Sale Amount
3. In the Custom column formula box, clear the existing = and type the formula below.

4. Click OK.
5. With the new Sale Amount column selected, click the type icon and set it to Decimal Number.

Custom Column Formula — Sale Amount

= [Quantity] * [Unit Price] * (1 - [Discount])

Formula Tips

Column names are case-sensitive and must match exactly — use the Available columns list on the right side of the Custom Column dialog to insert them without typos.

Wrap column names in square brackets [].

Basic operators: + (add), - (subtract), * (multiply), / (divide).

Expected Result

A new Sale Amount column appears at the far right of the table.

Sample values:

Keyboard × 12 units × \$54.99 × (1 - 0.00) = \$659.88

Bookshelf × 18 units × \$189.99 × (1 - 0.00) = \$3,419.82

File Cabinet × 20 units × \$239.99 × (1 - 0.20) = \$3,839.84

Task 5B — Add Discount Amount

Discount Amount shows how much was saved on each transaction — useful for analysing promotion effectiveness.

1. In the Add Column ribbon, click Custom Column again.
2. Set the column name to: Discount Amount
3. Enter the formula below and click OK.
4. Set the data type to Decimal Number.

Custom Column Formula — Discount Amount

= [Quantity] * [Unit Price] * [Discount]

Expected Result

A new Discount Amount column appears after Sale Amount.

When Discount = 0.00, Discount Amount = 0.

When Discount = 0.20 and UnitPrice = 239.99 and Quantity = 20: Discount Amount = \$959.96

9. Close & Apply

You have completed all five transformations. Now load the clean data into the Power BI data model.

1. Review the Applied Steps pane on the right. You should see steps including: Source, Promoted Headers, Changed Type, Renamed Columns, Removed Columns, and your Added Custom columns.
2. In the Home ribbon of the Power Query Editor, click Close & Apply.
3. Power BI loads the data and returns you to the Report view.
4. Click the Data icon (table symbol) on the left navigation bar.
5. Select SalesData from the Fields pane to see your transformed table.

Checkpoint — Final Table Should Have

11 columns: Transaction ID, Date, Customer Name, City, Category, Product, Quantity, Unit Price, Discount, Sales Rep, Region, Sale Amount, Discount Amount

Wait — that's 13! Count yours and make sure all 13 are present.

50 rows of data.

No errors (red exclamation marks) in any column.

10. Solutions

The M code below shows the complete Applied Steps for the SalesData query. You can view your own M code at any time in Power Query Editor by clicking Home > Advanced Editor.

Complete M Code — SalesData Query

Solution — Full M Query

```
let
    // Step 1: Connect to the CSV file
    Source = Csv.Document(
        File.Contents("C:\\Users\\YourName\\Downloads\\Retail_Sales_Q1_2024.csv"),
        [Delimiter=",", Columns=13, Encoding=1252, QuoteStyle=QuoteStyle.None]
    ),

    // Step 2: Use first row as column headers
    PromotedHeaders = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),

    // Step 3: Change data types
    ChangedType = Table.TransformColumnTypes(PromotedHeaders, {
        {"TransactionID", type text},
        {"Date", type date},
        {"CustomerName", type text},
        {"City", type text},
        {"State", type text},
        {"Category", type text},
        {"Product", type text},
        {"Quantity", Int64.Type},
        {"UnitPrice", type number},
        {"Discount", type number},
        {"SalesRep", type text},
        {"Region", type text},
        {"Notes", type text}
    }),

    // Task 1: Rename the query → done via the Queries pane (no M step generated)

    // Task 2: Remove columns
    RemovedColumns = Table.RemoveColumns(ChangedType, {"Notes", "State"}),

    // Task 3: Rename columns
    RenamedColumns = Table.RenameColumns(RemovedColumns, {
        {"TransactionID", "Transaction ID"},
        {"CustomerName", "Customer Name"},
        {"UnitPrice", "Unit Price"},
        {"SalesRep", "Sales Rep"}
    }),

    // Task 4: Change data types (Date, Quantity, Unit Price, Discount)
    // (Power BI may combine this with the initial ChangedType step above)

    // Task 5A: Add Sale Amount calculated column
```

```

AddedSaleAmount = Table.AddColumn(RenamedColumns, "Sale Amount",
    each [Quantity] * [Unit Price] * (1 - [Discount]),
    type number),

// Task 5B: Add Discount Amount calculated column
AddedDiscountAmount = Table.AddColumn(AddedSaleAmount, "Discount Amount",
    each [Quantity] * [Unit Price] * [Discount],
    type number)

in
    AddedDiscountAmount

```

Task-by-Task Quick Reference

Task	What You Did
1 — Rename Query	Queries pane → double-click → type SalesData → Enter
2 — Remove Columns	Right-click column header → Remove (for Notes and State)
3 — Rename Columns	Double-click column header → type new name → Enter (4 columns)
4 — Change Data Types	Click type icon left of column name → select correct type (4 columns)
5A — Sale Amount	Add Column ribbon → Custom Column → = [Quantity]*[Unit Price]*(1-[Discount])
5B — Discount Amount	Add Column ribbon → Custom Column → = [Quantity]*[Unit Price]*[Discount]

11. Reflection Questions

Answer the following questions based on your experience in this exercise.

1. Why is it important to click Transform Data instead of Load when connecting to a data source in Power BI?

2. What is the Applied Steps pane and why is it useful? What would happen if you deleted a step?

3. You removed the State column in Task 2. Can you think of a situation where keeping State might be valuable? What would you need to change in the exercise?

4. In Task 4, what problems could occur in your report if you left Quantity as a Text data type instead of Whole Number?

5. In Task 5, you created Sale Amount using a custom column formula. Why is it better to do this calculation in Power Query rather than in Excel before importing?

6. Look at the Applied Steps pane after completing all tasks. If a colleague sent you the same CSV with 100 new rows, what would you need to do to update the report?

Suggested Answers

1. Transform Data vs. Load

Clicking Load imports data immediately with Power BI's auto-detected types and no cleanup. Clicking Transform Data opens Power Query Editor so you can fix types, rename columns, remove junk data, and add calculations before the data enters the model. Starting clean saves correction work later.

2. Applied Steps pane

Applied Steps records every transformation as an ordered list of steps. You can click any step to preview the data at that point in time, and delete a step to undo that transformation. Deleting an early step can cause downstream steps to fail if they reference a column the deleted step created.

3. Keeping the State column

State would be valuable for geographic drilldowns (e.g. filtering sales by state on a map). You would keep the column, set its type to Text, and potentially use it to create a State / City hierarchy in the report.

4. Wrong data type for Quantity

If Quantity is Text, Power BI cannot sum, average, or multiply it. Aggregate measures (Total Units Sold, Sale Amount) would fail or produce errors. Sorting would be alphabetical (1, 10, 11, 2 ...) instead of numeric.

5. Calculating in Power Query vs. Excel

Power Query calculations are repeatable and automatic — every time the CSV is refreshed the formula re-runs. An Excel pre-calculation is manual and error-prone; it must be redone every time new data arrives. Power Query also keeps the original columns intact so the logic is transparent and auditable.

6. Refreshing with new data

Nothing — just replace the CSV file with the updated one (same name, same path) and click Refresh in Power BI. All Applied Steps run automatically on the new rows. This is the power of a repeatable ETL pipeline.

12. Challenge Extension

Finished early? Take your skills further with these challenges. Each builds on the transformations you completed above.

A	Add a Profit Margin % column. Assume a fixed cost of 60% of Unit Price for every product. The formula is: $= ([\text{Unit Price}] * 0.40 * [\text{Quantity}]) / [\text{Sale Amount}] * 100$ Format the result as a Decimal Number.
B	Add a Price Tier conditional column with these rules: • If Unit Price < 10 → "Budget" • If Unit Price < 50 → "Mid-Range" • Else → "Premium" (Use Add Column > Conditional Column — no formula needed.)
C	The SalesRep and Region columns always go together (one rep = one region). Create a Combined Rep-Region column that looks like this: Jordan Lee (East) Formula: $= [\text{Sales Rep}] \& " (" \& [\text{Region}] \& ") "$
D	Reorder the columns so they appear in this logical order: Transaction ID Date Customer Name City Category Product Quantity Unit Price Discount Discount Amount Sale Amount Price Tier Sales Rep Region (Hint: Transform ribbon > Reorder Columns, or drag column headers.)
E	Explore the Applied Steps pane. Click each step one by one starting from Source. Write a one-sentence description of what each step does in your own words.

Challenge Solutions

Challenge A — Profit Margin %

Custom Column Formula

```
= ([Unit Price] * 0.40 * [Quantity]) / [Sale Amount] * 100
```

Challenge B — Price Tier (Conditional Column)

```
// Configured via the Conditional Column dialog - M code generated:  
= Table.AddColumn("#...", "Price Tier",  
    each if [Unit Price] < 10 then "Budget"  
        else if [Unit Price] < 50 then "Mid-Range"  
        else "Premium",  
    type text)
```

Challenge C — Combined Rep-Region

Custom Column Formula

```
= [Sales Rep] & " (" & [Region] & ") "
```